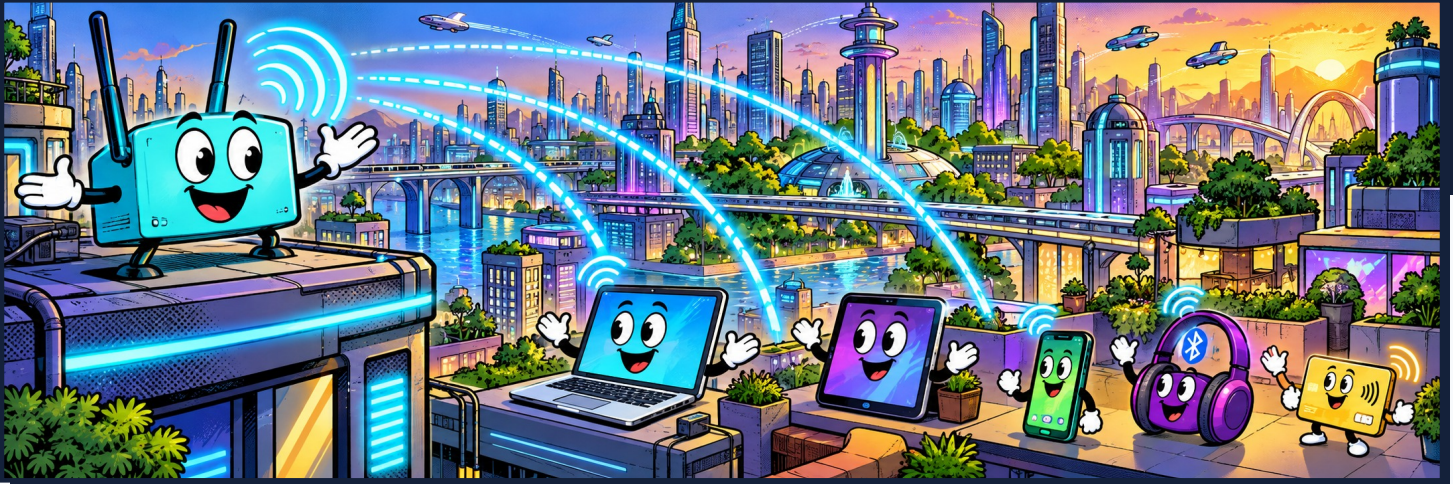


NETWORK CITY

WI-FI & WIRELESS

Standards, signals, and memory
practice



LEARN THE BAND. RECALL THE STANDARD. SOLVE THE SCENARIO.

Detailed comic illustrations + short definitions + clear exam clues

Explore Wi-Fi standards, bands, channels, Bluetooth, NFC, and RFID. Learn the terms, then practice 24 flashcards in the matching clickable PowerPoint. Includes printable card pairs and a three-question quiz.

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- 03 Radio words: frequency, band & channel
- 04 Three bands: different tradeoffs
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- 06 Widths, radar & radio rules
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Independent study guide. Aligned to exam 220-1201; not an official CompTIA publication. Illustrations are memory aids. Technical notes and labeled diagrams explain the real rules.

Meet the wireless crew

VISUAL: Draw a cyan access point sending radio arcs to a laptop, tablet, and phone across a neon city.



WOOSH! Our messages travel through the air!

WOOSH!

- **Wireless.** A connection that sends information without a cable between the communicating devices. This guide focuses on radio links.

- **Wi-Fi.** A brand name for wireless networking based on the IEEE 802.11 family. Learn Wi-Fi as a name, rather than an acronym to expand.

- **IEEE.** Institute of Electrical and Electronics Engineers. This organization develops standards, including the 802.11 family for wireless local networks.

- **WLAN / LAN.** Wireless Local Area Network / Local Area Network. A local network connects nearby devices, such as those in a home or school.

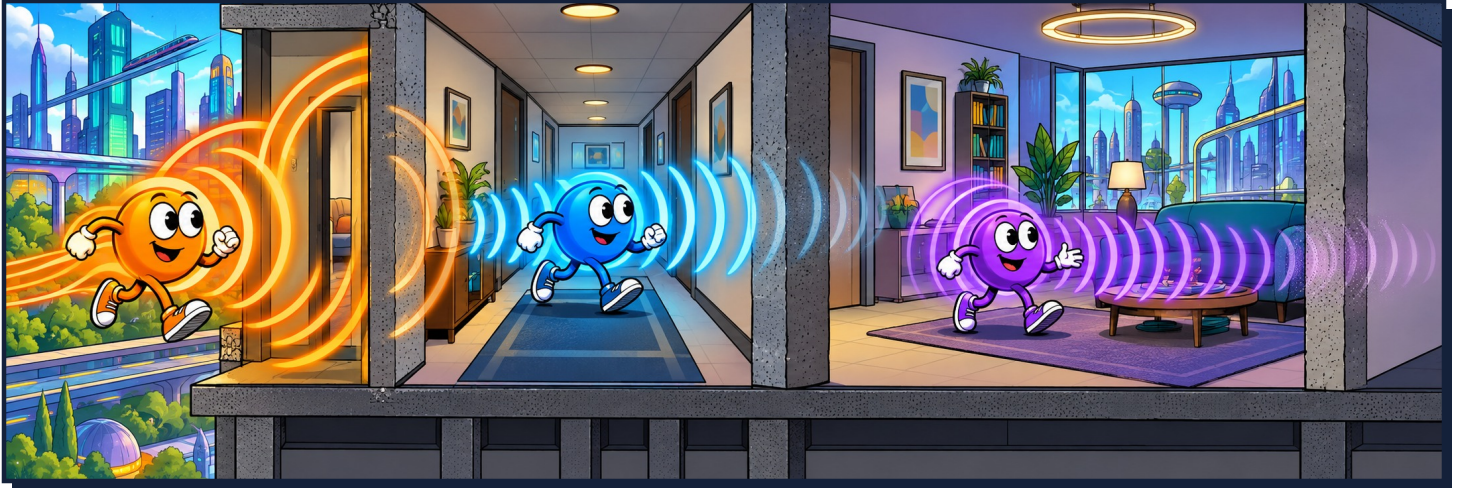
- **AP / client.** An Access Point connects Wi-Fi clients to a network. A client is a device using that connection. A home wireless router often combines an AP, router, and Ethernet switch.

- **Internet access.** Wi-Fi supplies a local link. An ISP, or Internet Service Provider, supplies an internet connection. You can have working Wi-Fi while the internet connection is down.

EXAM MAP: Core 1 (220-1201), objective 2.2: bands, channels, regulations, 802.11, Bluetooth, NFC, and RFID. Setup and troubleshooting chapters add related context. [1]

Radio words: frequency, band & channel

VISUAL: Draw orange, blue, and purple radio couriers moving through different rooms in a bright cutaway building.



Pick the band. Find the channel!

TUNE!

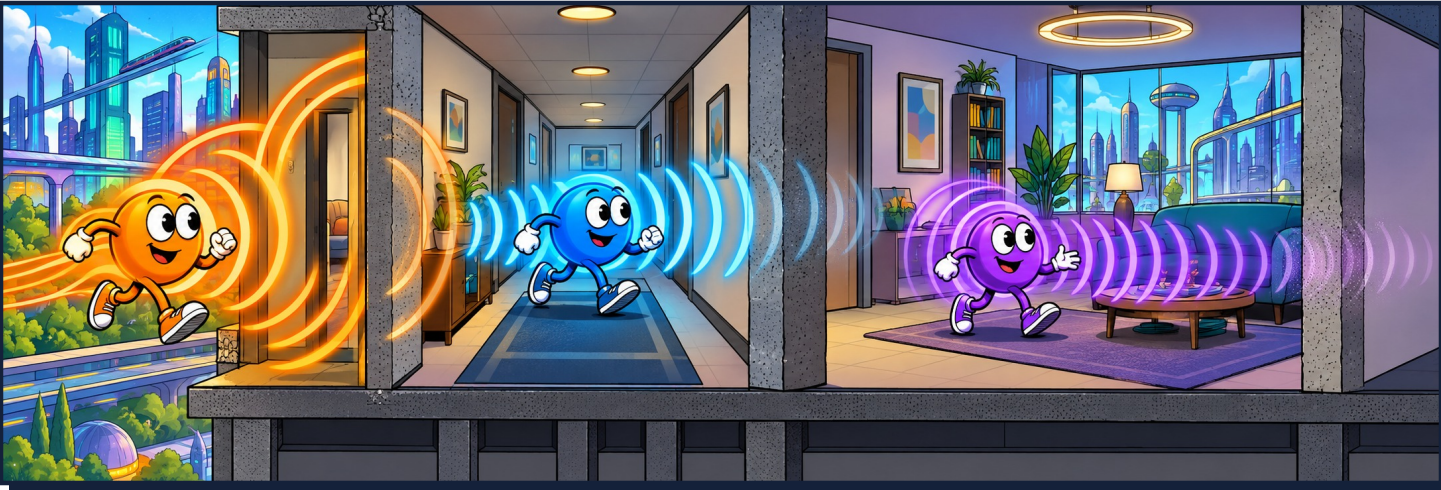
- **RF - Radio Frequency.** The part of the electromagnetic spectrum used for radio communication. Wi-Fi radios turn digital data into changing radio signals.
- **Frequency.** How often a wave repeats each second. Hz means hertz: cycles per second. MHz means millions of cycles. GHz means billions of cycles.
- **Band.** A range of radio frequencies. Picture a neighborhood. Common Wi-Fi band names are 2.4 GHz, 5 GHz, and 6 GHz.

- **Channel.** A smaller slice of a band. Picture a street inside that neighborhood. Nearby networks share the available radio space.
- **Channel width.** How much frequency space a channel uses, measured in MHz. Picture the street's width. Wider channels can carry more data when conditions and devices support them.
- **Data rate.** How quickly bits travel. Mb/s means megabits per second. Gb/s means gigabits per second. These units describe data, while GHz describes a radio frequency.

NUMBER TRAP: 5 GHz is a radio band. Wi-Fi 5 is a generation name. Cellular 5G is a mobile-network generation. The shared number does not make them the same thing.

Three bands: different tradeoffs

VISUAL: Draw orange waves crossing a doorway, blue waves in a hallway, and purple waves in a clear room.



Walls, neighbors, and devices change the ride!

ZOOM!

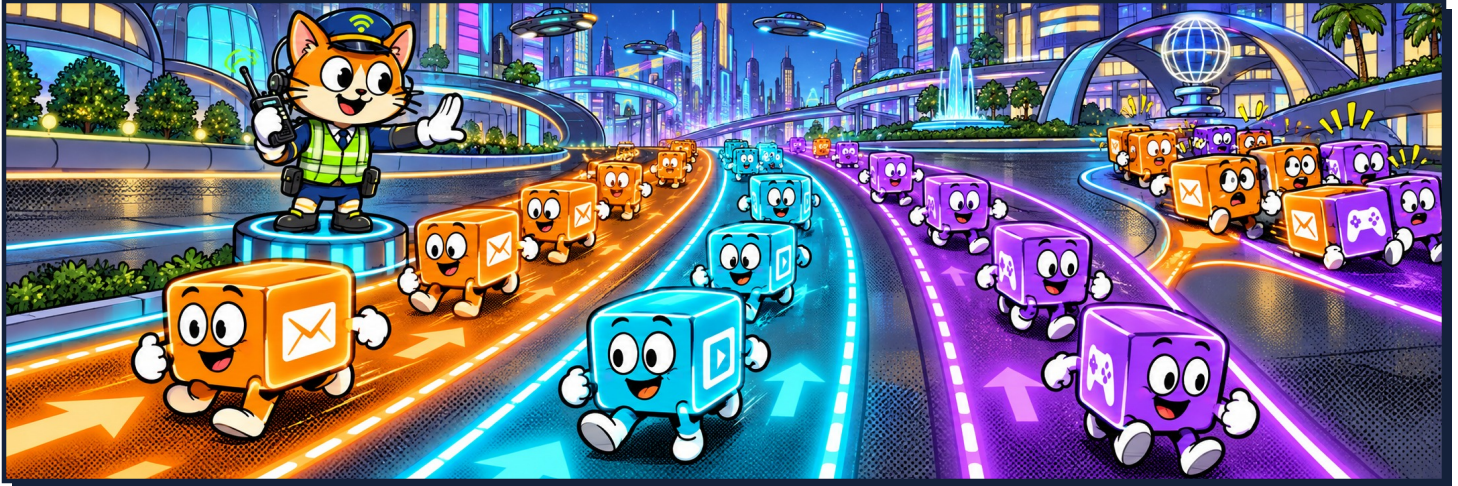
Band	Useful strengths	Watch for
2.4 GHz	Often better reach through walls; many older and IoT devices	Crowded spectrum; fewer separate channels
5 GHz	More channel choices; often useful for faster nearby links	Walls and distance can weaken the link
6 GHz	Additional spectrum for compatible Wi-Fi 6E/7 devices	Needs supported hardware, software, security, and local permission

- IoT - Internet of Things.** Connected objects such as sensors, lights, and smart plugs. Many simple devices support only 2.4 GHz.
- Reach is conditional.** At similar power and other conditions, lower frequencies often handle walls better. Building materials, antennas, rules, and receiver quality also matter.
- Use case.** A nearby fast laptop may suit 5 or 6 GHz. A basic sensor farther away may suit 2.4 GHz. Test at the actual location. [2,3]

CHECK BOTH ENDS: A tri-band access point does not add a 6 GHz radio to an older laptop. The AP and client must share a supported band and mode.

Channels: the 1, 6, 11 memory hook

VISUAL: Draw a radio traffic marshal separating orange, cyan, and purple lanes. Overlapping lanes cause a small jam.



Three clear lanes beat a tangled pile!

BEEP!

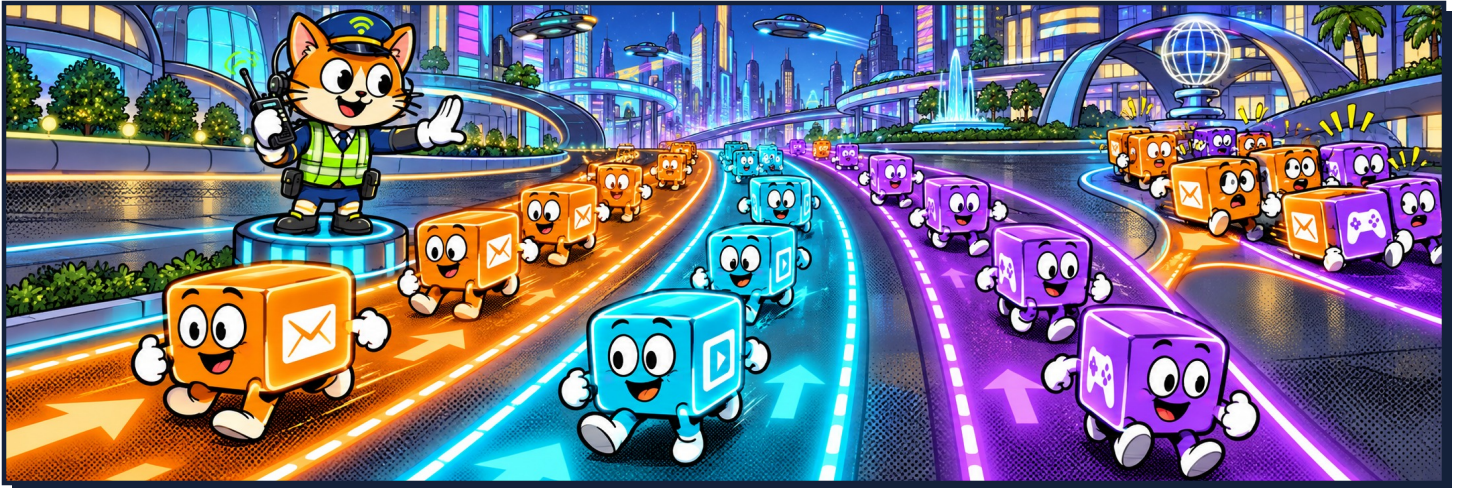
- **The common U.S. plan.** For 2.4 GHz Wi-Fi using 20 MHz channels, remember channels **1, 6, and 11**. They form the familiar nonoverlapping plan. [4]
- **Overlap.** Nearby channel numbers can occupy some of the same frequency space. Choosing channel 2 beside channel 1 does not create a clean extra lane.
- **Same-channel sharing.** Nearby networks on the same channel compete for airtime. They may hear each other and take turns. A shared channel can still become busy.

- **Adjacent-channel interference.** Overlapping radio activity can disrupt transmissions. Devices may need to retry data, which uses more airtime.
- **Channel selection.** Use approved automatic channel planning or survey the area. Pick a suitable channel and width, then test performance. A quiet-looking channel can change later.
- **Regional rules.** The permitted channel list depends on location. This guide uses U.S. examples. Keep the device's country or region setting correct.

ROTE MEMORY: Say “2.4, twenty wide, one-six-eleven.” This hook describes the common U.S. 2.4 GHz plan, not every band or every country.

Widths, radar & radio rules

●●● **VISUAL:** Draw wide and narrow neon lanes beside a careful radio marshal holding a blank permission card.



More lanes help only when the road is clear!

CHECK!

- **Common widths.** Wi-Fi modes may use 20, 40, 80, or 160 MHz channels. Wi-Fi 7 adds support for 320 MHz in 6 GHz where allowed. Both endpoints must support the chosen mode. [3,5]

- **The tradeoff.** A wider channel occupies more spectrum. In a crowded building, a narrower channel may leave more usable channels and perform better overall.

- **DFS - Dynamic Frequency Selection.** Some 5 GHz channels share spectrum with radar. On DFS channels, equipment must detect radar and follow required channel-avoidance procedures. A channel change can interrupt service. [2]

- **FCC.** Federal Communications Commission, the U.S. radio regulator. Unlicensed spectrum still has rules for power, equipment, location, and use.

- **6 GHz example.** U.S. standard-power operation uses AFC, Automated Frequency Coordination, to help protect existing licensed links. LPI means Low Power Indoor. Device classes have different rules. [6]

- **Safe setup.** Use approved equipment, the correct regional settings, and the manufacturer's placement and exposure guidance. Do not defeat radio limits or use an indoor-only unit outdoors.

EXAM CLUE: Frequency, channel number, and channel width describe different things. Local rules determine which combinations your equipment may use.

The classic crew: 802.11a, b & g

VISUAL: Draw a gold bicycle courier beside blue and green sedan couriers in a neon wireless museum.



Old names still show up in exam clues!

VROOM!

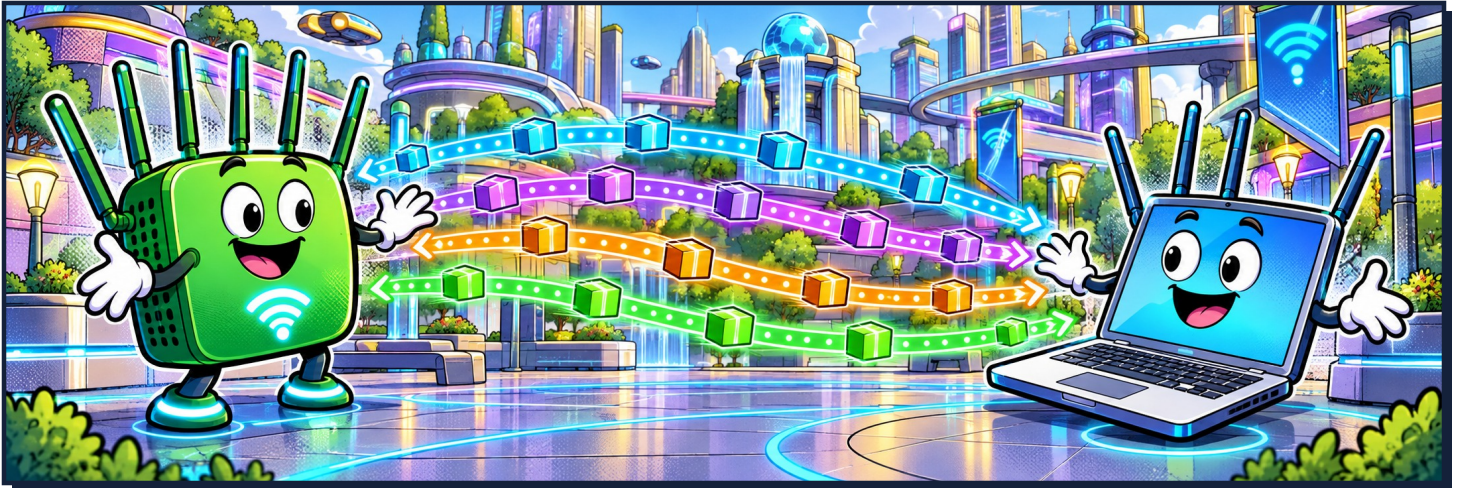
Standard	Band	Maximum PHY rate	Memory hook
802.11a	5 GHz	54 Mb/s	A takes the 5 GHz road
802.11b	2.4 GHz	11 Mb/s	B is the bicycle: 11
802.11g	2.4 GHz	54 Mb/s	G stays at 2.4, goes 54

- PHY - Physical layer.** The radio link’s signaling layer. These are theoretical maximum link rates under supported conditions. Actual file transfers run slower. [2,7]
- The same-speed trap.** 802.11a and 802.11g share a 54 Mb/s maximum but use different bands. 802.11b and g share 2.4 GHz but have different maximum rates.
- Use cases.** Recognize legacy devices and compatibility clues. Old clients can add airtime overhead. Support depends on the AP’s enabled modes and security.

SAY IT: A = 5 / 54. B = 2.4 / 11. G = 2.4 / 54. Band first, rate second. We use the standard names rather than unofficial retroactive Wi-Fi 1/2/3 labels.

Wi-Fi 4: 802.11n adds more paths

VISUAL: Draw a green access point and laptop exchanging several separate colored streams across a neon courtyard.



N brings multiple paths to the team!

MULTI!

- **Name and bands.** Wi-Fi 4 means 802.11n. The standard supports 2.4 GHz and 5 GHz. A specific product may implement only one band.

- **Top rate.** Up to 600 Mb/s under the maximum four-stream, 40 MHz, short-guard-interval configuration. Smaller client configurations have lower ceilings. [7]

- **MIMO.** Multiple Input, Multiple Output. Several transmit and receive paths can improve wireless performance. Compatible devices can carry separate spatial streams.

- **Spatial stream.** An independent data stream sent through the radio channel. More streams require support at both ends. Count supported streams, not just visible antennas.

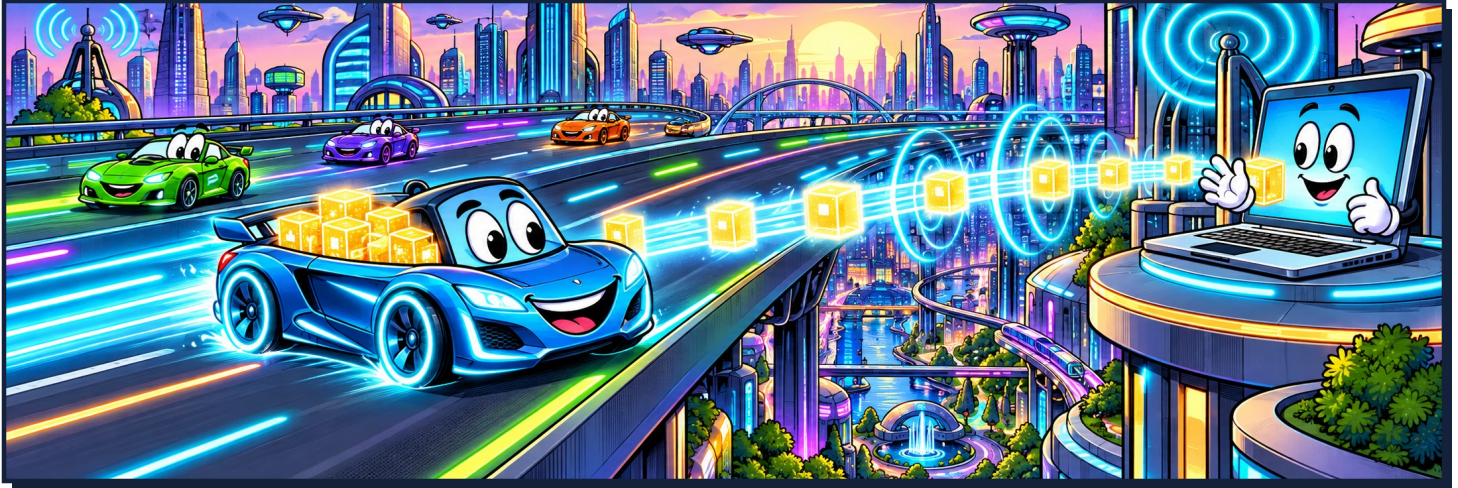
- **Width.** 802.11n supports 20 and 40 MHz operation. A 40 MHz channel uses more spectrum. The 2.4 GHz band can become crowded quickly. [4]

- **Use case.** Identify older dual-band equipment and devices that need basic wireless access. Check the actual adapter, AP, channel width, and security support.

MEMORY HOOK: N = Wi-Fi 4. Two possible bands. Up to 600 Mb/s. A 600 Mb/s standard ceiling does not make every Wi-Fi 4 device a 600 Mb/s device.

Wi-Fi 5: 802.11ac takes the fast road

VISUAL: Draw a blue sports-car courier on a wide elevated radio road leading to a laptop terminal.



AC cruises on the 5 GHz road!

ZOOM!

- **Name and band.** Wi-Fi 5 means 802.11ac. Its radio operation uses the 5 GHz band. A dual-band router labeled AC usually uses another standard on its 2.4 GHz radio.

- **Top rate.** About 6.9 Gb/s at the standard's theoretical maximum configuration. Ordinary phones and laptops support fewer streams and much lower link ceilings. [2,8]

- **Wider channels.** The standard supports widths up to 160 MHz, including supported 80+80 MHz arrangements. Product and deployment support varies.

- **MU-MIMO.** Multi-User Multiple Input, Multiple Output. Supported equipment can send separate spatial streams to multiple compatible clients at once. Wi-Fi 5 introduced downlink MU-MIMO.

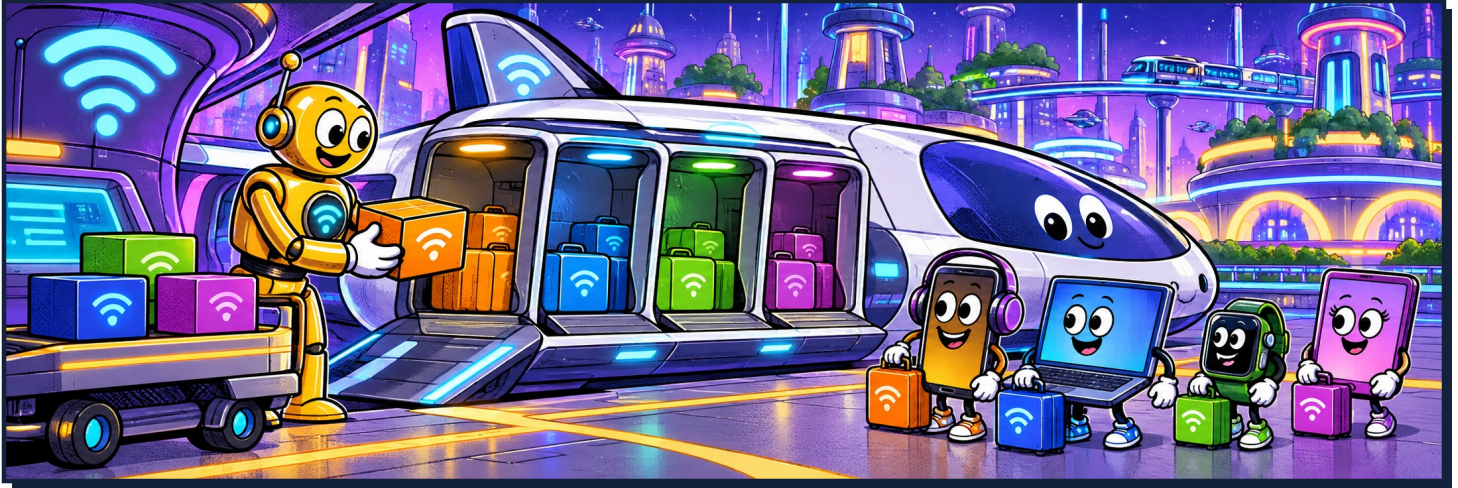
- **Beamforming.** A transmitter adjusts signals across antennas to improve delivery toward a receiver. Picture a focused voice. It cannot make solid walls disappear.

- **Use case.** Recognize common 5 GHz equipment and fast local-network links. Check the client's stream count and channel support before predicting performance.

MEMORY HOOK: AC = Wi-Fi 5 = 5 GHz. Maximum standard rate about 6.9 Gb/s. Product labels often add several radios together; they are not one client's speed.

Wi-Fi 6: 802.11ax shares airtime

VISUAL: Draw a gold dispatcher putting small parcels for several device characters into sections of one shuttle.



One planned trip can serve several devices!

LOAD!

- **Name and bands.** Wi-Fi 6 means 802.11ax in 2.4 and 5 GHz. Its biggest benefits include more efficient service when many devices share the air.

- **Top rate.** About 9.6 Gb/s at the standard's maximum configuration. This is a theoretical PHY ceiling, not a promised internet download rate. [2]

- **OFDMA.** Orthogonal Frequency Division Multiple Access. The AP can assign small portions of a channel to different users within one transmission. Think of separate seats on a shared shuttle.

- **Resource unit.** An assigned group of radio subcarriers inside the channel. OFDMA uses these smaller pieces to serve compatible devices efficiently.

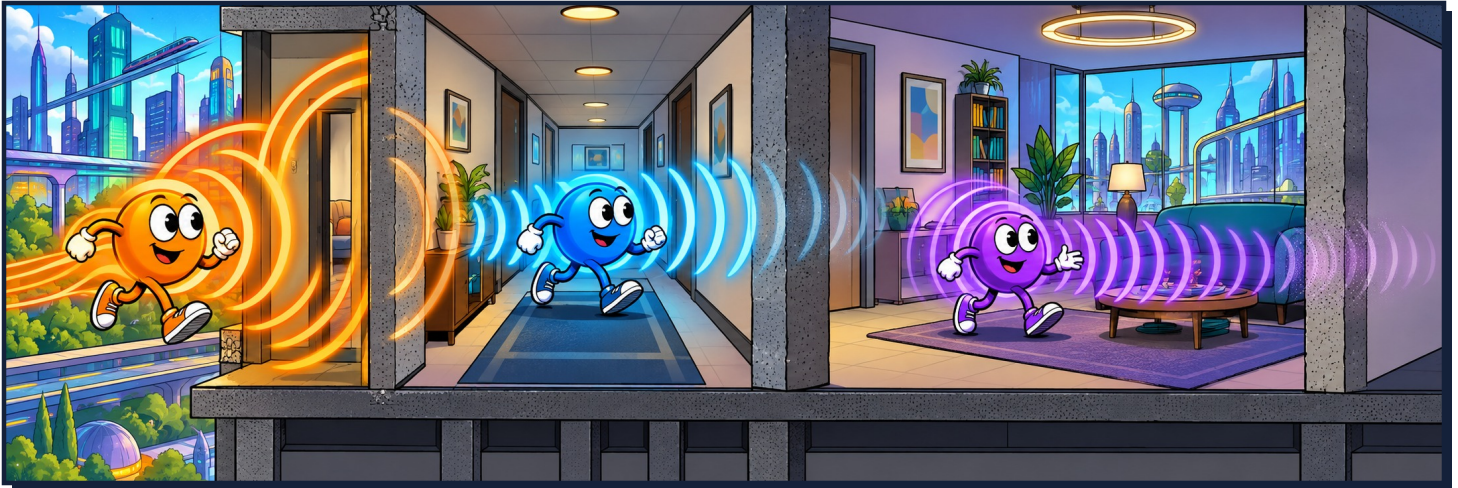
- **TWT - Target Wake Time.** Devices can arrange wake times for communication. Compatible clients may save battery power by sleeping between scheduled activity.

- **Use case.** Busy classrooms, offices, or homes with many active clients. Wi-Fi 6 also supports uplink and downlink multi-user features. Benefits depend on device support and traffic. [9]

MEMORY HOOK: AX = Wi-Fi 6. OFDMA helps share channel space. MIMO uses multiple spatial paths. The two ideas can work together.

Wi-Fi 6E: the extra 6 GHz neighborhood

●●● **VISUAL:** Draw the purple radio courier entering a spacious new room while orange and blue couriers use older rooms.



E extends Wi-Fi 6 into extra spectrum!

EXTRA!

- **6E meaning.** Wi-Fi 6E extends Wi-Fi 6 technology into the 6 GHz band. It still uses 802.11ax. E is not a separate IEEE amendment. [3]

- **Why it matters.** Extra spectrum can provide more room for wide channels. Older 2.4/5 GHz-only clients cannot transmit on this band, which changes the mix of competing devices.

- **Same ceiling family.** Wi-Fi 6E does not change the 802.11ax maximum PHY ceiling of about 9.6 Gb/s. Cleaner spectrum can still improve real results.

- **Compatibility chain.** Check the client radio, AP, operating-system support, driver, security mode, and local 6 GHz rules. A Wi-Fi 6 label alone does not promise 6 GHz support.

- **Security.** 6 GHz Wi-Fi uses newer security requirements. WPA3 protects authenticated networks; Enhanced Open uses OWE, Opportunistic Wireless Encryption, for encrypted open access. Legacy WPA2/open modes do not carry over unchanged. [10]

EXAM TRAP: Wi-Fi 6 = generation. 6 GHz = band. Wi-Fi 6E = Wi-Fi 6 technology extended into 6 GHz. Match the full name before choosing an answer.

Wi-Fi 7: extra context for newer gear

VISUAL: Draw a purple express courier using two glowing paths between the same access point and laptop.



Two supported links can work as a team!

LINK!

- **Name and bands.** Wi-Fi 7 means 802.11be. The technology spans 2.4, 5, and 6 GHz, subject to the device and local rules.

- **Top rate.** About 46.1 Gb/s is the standard's theoretical maximum using its largest supported configuration. Most devices support far fewer streams. This number is not a normal laptop download speed. [2]

- **320 MHz.** A new wide-channel option in 6 GHz where permitted. A device with Wi-Fi 7 branding may still have a narrower maximum channel width.

- **MLO - Multi-Link Operation.** Compatible devices can coordinate multiple links. Supported modes can improve throughput, reliability, or delay. Implementation details matter. [5]

- **QAM.** Quadrature Amplitude Modulation. It encodes information using changes in a signal's amplitude and phase. Wi-Fi 7 supports 4096-QAM; higher-order modes need good signal conditions.

- **Use case.** Recognize newer equipment and naming. Master the common A+ band/channel concepts before memorizing advanced feature details.

SCOPE NOTE: Objective 2.2 names 802.11 standards broadly. Wi-Fi 7 detail is included as clearly marked supporting context, not a prediction of a specific exam question.

The standards memory map

●●● **VISUAL:** Draw the classic vehicle couriers beside the green multi-path team and purple express courier in a neon parade.



Name it. Band it. Rate it!

RECALL!

Standard	Generation	Band(s), GHz	Max PHY rate*
802.11a	Legacy	5	54 Mb/s
802.11b	Legacy	2.4	11 Mb/s
802.11g	Legacy	2.4	54 Mb/s
802.11n	Wi-Fi 4	2.4 / 5	600 Mb/s
802.11ac	Wi-Fi 5	5	About 6.9 Gb/s
802.11ax	Wi-Fi 6	2.4 / 5	About 9.6 Gb/s
802.11ax	Wi-Fi 6E	Adds 6	About 9.6 Gb/s
802.11be	Wi-Fi 7	2.4 / 5 / 6	About 46.1 Gb/s

***Read the ceiling correctly:** These are theoretical standard maximums, using the maximum supported streams, width, and other settings. Actual products, negotiated links, and useful throughput are lower or different. [2,7,8]

REPEAT FROM MEMORY: a, b, g, n, ac, ax. Then say the band and maximum rate. Add 6E as the 6 GHz extension and be as Wi-Fi 7 context.

Speed labels versus actual downloads

VISUAL: Draw the blue courier waiting at a narrow delivery gate after leaving a wide fast radio road.



The narrowest busy part slows the trip!

WAIT!

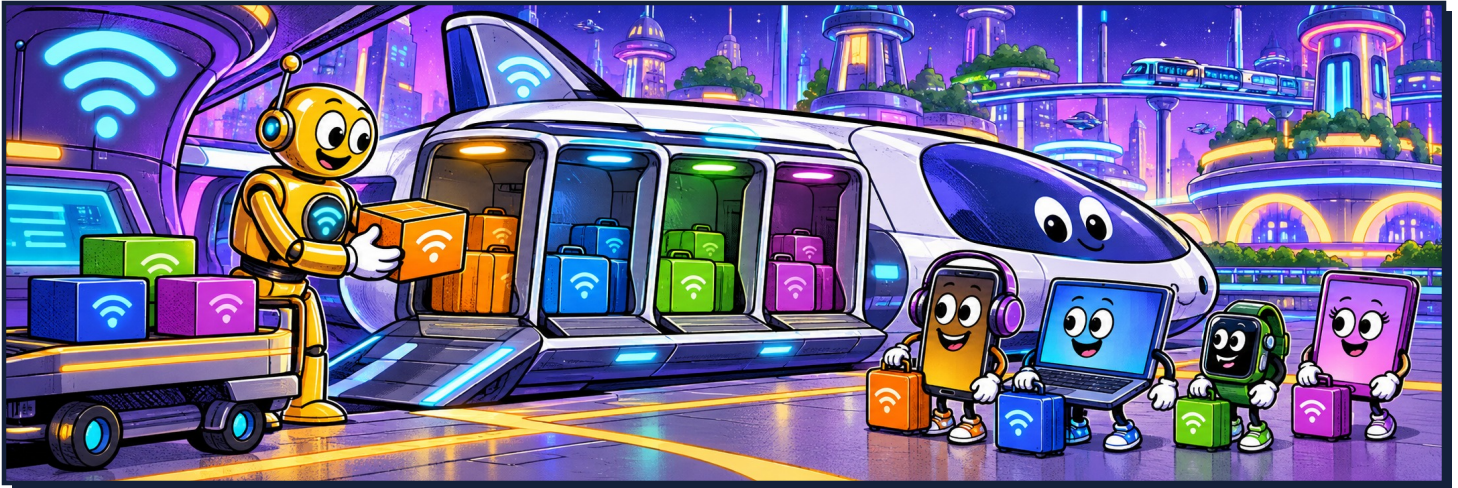
- **Link rate.** The radio speed that the AP and client negotiate. It changes with the signal and supported settings. It includes more than the useful file data.
- **Throughput.** Useful data delivered over time. Headers, acknowledgments, waiting, retries, and other users reduce the useful result.
- **Bottleneck.** The part limiting the whole trip. Your internet plan, AP uplink, server, VPN, storage, or client can limit a download even when the radio link is fast. VPN means Virtual Private Network.

- **Mb/s versus MB/s.** A lowercase b means bits. An uppercase B means bytes. Eight bits make one byte. A 100 Mb/s connection has a raw arithmetic ceiling of 12.5 MB/s before overhead.
- **Latency.** Delay, often measured in milliseconds (ms). Low delay helps calls and games. A high maximum data rate does not guarantee low delay.
- **Test fairly.** Compare a local transfer with an internet test. Check the negotiated band, width, and rate. Test near the AP and at the problem spot. [11]

EXAM CLUE: A router's combined marketing number may add multiple radios. It does not describe one client's usable speed. Read the per-radio and per-client specifications.

Airtime: taking turns and sharing

VISUAL: Draw a radio dispatcher loading a shared shuttle while nearby device characters wait for their turn.



Listen first. Take a turn. Retry when needed!

SHARE!

- **Airtime.** Time spent using the radio channel. Many devices can share a network, but they compete for finite radio resources.

- **CSMA/CA.** Carrier Sense Multiple Access with Collision Avoidance. Wi-Fi devices listen and use waiting rules to reduce collisions. Wireless collisions and lost frames can still happen.

- **Acknowledgment / retry.** A receiver can confirm delivery. If confirmation fails, the sender may try again. Repeated retries use time that could carry new data.

- **MIMO versus OFDMA.** MIMO uses multiple spatial paths. OFDMA divides channel resources among users. Think parallel delivery paths versus assigned seats in a shuttle.

- **MU-MIMO.** Multi-user MIMO sends different spatial streams for multiple clients. Ordinary MIMO does not automatically mean several users receive at once.

- **Why busy rooms slow down.** Many active clients, weak links, interference, and management traffic all consume airtime. A strong signal alone does not prove the channel is quiet. [9,11]

MEMORY HOOK: MIMO = paths. MU-MIMO = paths for multiple users. OFDMA = pieces of a channel for multiple users. All need compatible equipment and suitable conditions.

Range: a site survey beats a promise

VISUAL: Draw a robot checking a tablet in several rooms while radio light fades behind brick walls and furniture.



A distance number needs a real-world test!

SCAN!

- **No universal Wi-Fi range.** A standard name does not promise one fixed distance in feet. Power, antennas, walls, interference, band, and required data rate change usable coverage.
- **Attenuation.** Loss of signal strength. Distance and obstacles weaken signals. Concrete, metal, and water can cause large losses.
- **Signal and noise.** RSSI means Received Signal Strength Indicator. SNR means Signal-to-Noise Ratio. A receiver needs the wanted signal to stand out from unwanted radio energy.

- **dBm.** Decibels referenced to one milliwatt. On common Wi-Fi readings, -50 dBm is stronger than -75 dBm. These are example readings, not universal pass/fail thresholds.
- **Site survey.** Check coverage, interference, and performance where devices actually operate. Test both directions because a small client may transmit less strongly than the AP. [12]
- **Feet-based practice.** Test at 10 ft, 30 ft, and 60 ft, then behind a wall. These are suggested test locations, not guaranteed range ratings. Record the results and the room layout.

DISTANCE RULE: Use inches or feet for physical measurements. Keep GHz and MHz for radio frequencies. No fixed “Wi-Fi 6 reaches exactly ___ ft” claim is reliable everywhere.

Placement, antennas & interference

VISUAL: Draw a robot moving an access point away from metal furniture and a microwave toward an open central position.



A better spot can beat a louder signal!

MOVE!

- **Placement.** Put an AP where it can cover the intended users. Follow the mounting instructions. Avoid burying it in a metal cabinet or behind dense obstacles.
- **Omnidirectional antenna.** Spreads energy around an axis. The pattern is not a perfect ball. Orientation and the antenna design affect coverage.
- **Directional antenna.** Concentrates energy toward a chosen area. Useful for planned links or specific coverage zones. It needs correct alignment and an approved design.

• **Interference.** Unwanted radio activity. Nearby Wi-Fi, Bluetooth, and some appliances can compete with or disrupt 2.4 GHz operation. A Wi-Fi scanner does not detect every non-Wi-Fi source.

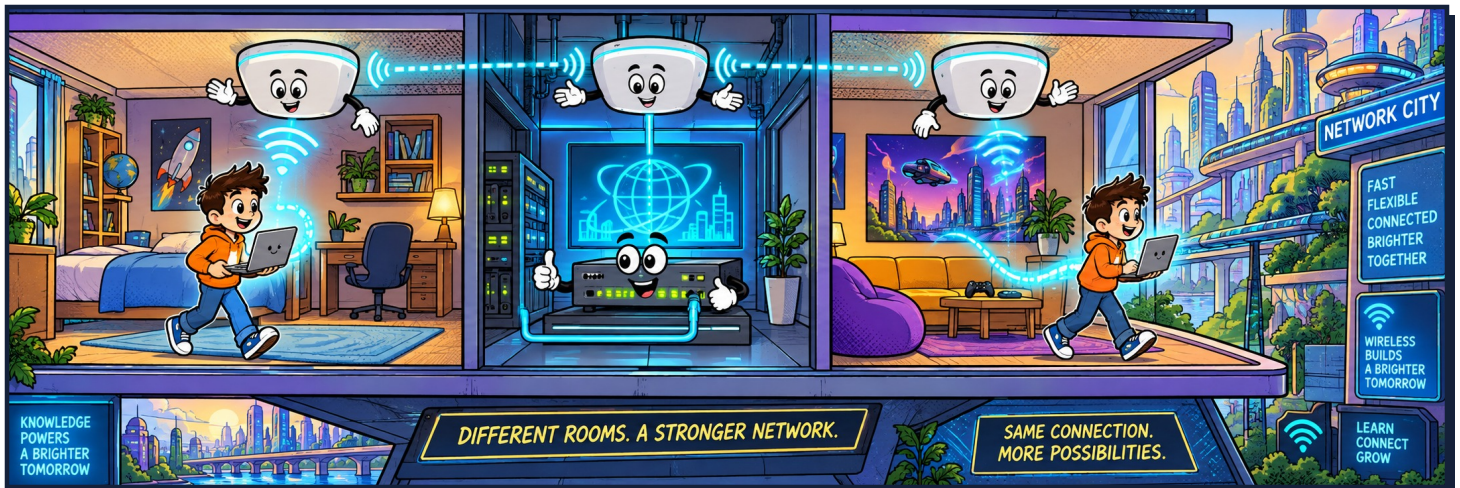
• **Power balance.** Turning AP power to maximum may create a one-way-looking link or more interference. Plan AP placement, channel use, and power together.

• **Use case.** A call works near the AP but fails in a back room. Compare the same device in both places, check obstacles and noise, then retest after one change. [11,12]

SAFE INSTALL: Follow mounting, cable, electrical, and RF-exposure instructions. Use outdoor-rated equipment where needed. Check coverage after furniture or walls change.

Network names, radios & roaming

VISUAL: Draw a laptop walking between rooms served by friendly access points with a continuous soft radio glow.



The network name stays familiar as I move!

ROAM!

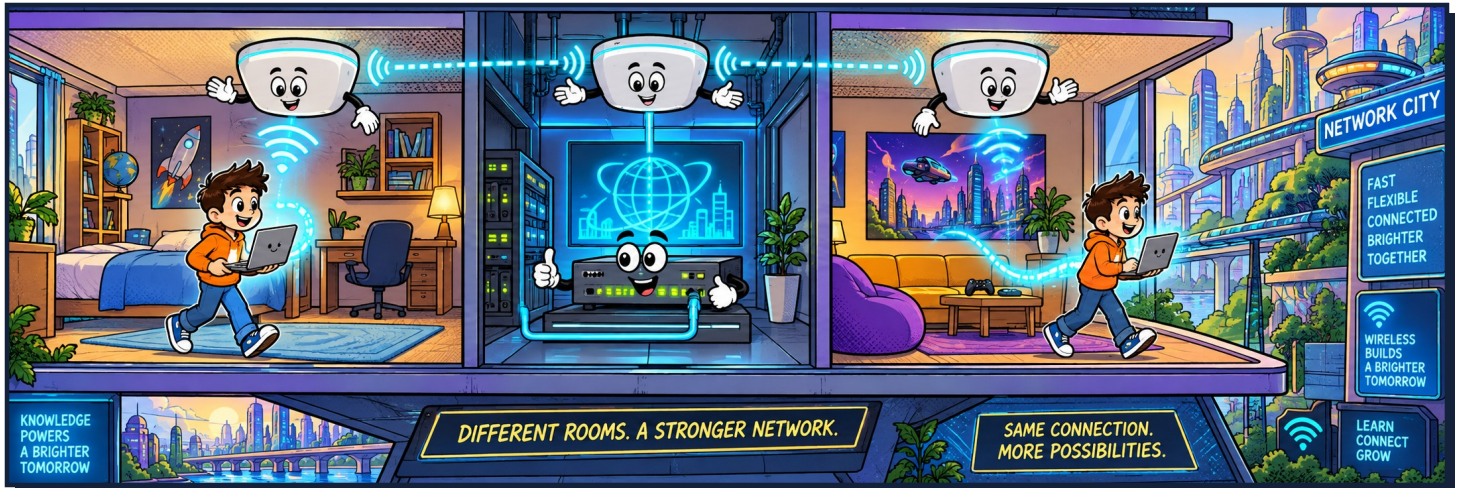
- **SSID.** Service Set Identifier. The network name users select. Several APs may advertise the same SSID as part of one planned network.
- **BSSID.** Basic Service Set Identifier. An identifier for a particular basic service set, commonly based on an AP radio interface's MAC address.
- **MAC address.** Media Access Control address. A link-layer identifier. Many client systems use private or changing Wi-Fi MAC addresses to reduce tracking.

- **Roaming.** A client changes from one AP to another. The client generally decides when to move; network features can assist. A shared name alone does not guarantee seamless calls.
- **Band steering.** An AP system can encourage a suitable client to use another band. The client must support that band, and actual behavior varies.
- **Hidden SSID.** Hiding the name does not make the network invisible or secure. Use supported authentication and encryption. [13,14]

EXAM HOOK: SSID = network name. BSSID = a particular wireless service set. Password = a credential. These three items do different jobs.

Mesh, hotspots & direct connections

VISUAL: Draw three access-point friends joining separate rooms through backhaul paths while one links to a wired hub.



Every wireless hop still needs a path home!

HOP!

- **Mesh Wi-Fi.** Several nodes cooperate to extend network coverage. A node needs a reliable route to the rest of the network.
- **Backhaul.** The connection carrying traffic between an AP or mesh node and the upstream network. It may use Ethernet or a wireless link.
- **Repeater / extender.** Receives and forwards wireless traffic to extend coverage. A shared-radio repeating hop can consume extra airtime. Results depend on the design.

- **Hotspot.** An access point offering network access. A phone hotspot usually shares cellular data over a local Wi-Fi link. Carrier limits and battery use still apply.
- **Tethering.** Sharing one device's internet connection with another device. Depending on support, the connection may use Wi-Fi, USB, or Bluetooth. USB means Universal Serial Bus.
- **Wi-Fi Direct.** Compatible devices form a direct Wi-Fi connection without joining a conventional AP-based WLAN. Support, discovery, and security depend on the devices. [15]

PLACEMENT CLUE: An extender in a dead zone may repeat a poor link. It needs a usable upstream connection. A stronger Wi-Fi icon does not fix a broken internet uplink.

Wireless security: keys and shields

VISUAL: Draw a purple shield gatekeeper checking devices at a city entrance with a separate guest gate and locked parcels.



Join the right network. Protect the message!

LOCK!

- **Authentication.** Checks whether a device or user may join. Encryption scrambles data so an unauthorized listener cannot simply read it.

- **WPA2 / WPA3.** Wi-Fi Protected Access 2 / 3. Use supported modern security. WPA2 with AES may support older devices; WPA3 provides newer protections. AES means Advanced Encryption Standard. [13]

- **Personal.** Common home mode. WPA2-Personal uses a PSK, Pre-Shared Key. WPA3-Personal uses SAE, Simultaneous Authentication of Equals, for password-based authentication.

- **Enterprise.** Typically uses individual credentials or certificates with 802.1X authentication. A RADIUS server commonly supports it. RADIUS means Remote Authentication Dial-In User Service.

- **Legacy traps.** WEP means Wired Equivalent Privacy. Original WPA and TKIP, Temporal Key Integrity Protocol, are also outdated. Recognize them in study questions, but avoid them in a new secure setup.

- **Guest access.** A guest network can isolate visitors from internal devices when configured correctly. A captive portal is a sign-in or terms page; the page alone does not prove radio encryption. [16]

SUPPORTING CONTEXT: Detailed security configuration extends beyond objective 2.2. Never treat a hidden name, MAC filter, or padlock-looking website as proof of a trustworthy wireless network.

Connection mission: join and verify

VISUAL: Draw a phone and laptop greeting a cyan access point while a technician checks a gold card.



Connected is the start. Working is the test!

JOIN!

- **1. Match support.** Confirm the band, Wi-Fi mode, security, and driver support. Turn on the Wi-Fi radio and check airplane-mode settings.
- **2. Choose the right SSID.** Confirm the expected network name with the owner or approved setup instructions. A familiar-looking name can be copied.
- **3. Authenticate.** Enter the correct passphrase or required enterprise credentials. Check certificates according to the organization's instructions.
- **4. Check IP settings.** DHCP means Dynamic Host Configuration Protocol. It commonly supplies IP settings automatically. Verify the address, gateway, and DNS settings.
- **5. Test the path.** Test the local network, then internet access and name lookup. DNS means Domain Name System. It translates names into addresses and other records.
- **6. Verify the real job.** Try the needed call, website, print job, or transfer. Record the settings and result. Restore only approved settings if a change fails.

LAYER CLUE: A Wi-Fi authentication failure differs from a DHCP failure, a DNS failure, or an ISP outage. Find the failing step before replacing hardware.

Bluetooth: the nearby-device team

VISUAL: Draw blue headphones, a mouse, keyboard, fitness sensor, and phone trading tiny signals around a bright desk.



Small devices can make a useful local team!

PAIR!

- **Bluetooth.** A wireless technology often used for nearby accessories, audio, sensors, and data links. The name is not an acronym.

- **Band.** Bluetooth operates in 2.4 GHz. It shares spectrum with other users, including 2.4 GHz Wi-Fi. It uses frequency hopping to move among channels. [17]

- **Classic / LE.** Bluetooth Classic includes BR/EDR: Basic Rate / Enhanced Data Rate. LE means Low Energy, often used for sensors and modern low-power features. LE also supports audio in compatible products.

- **PAN.** Personal Area Network. A network around a person and their devices, such as a phone and nearby accessories.

- **Range.** About 33 ft is a familiar short-range classroom example, not a universal limit. Bluetooth range varies greatly with the PHY, power, antennas, environment, and product. [18]

- **Profiles and roles.** A profile defines behavior for a job, such as audio or a keyboard. Two devices need compatible features and roles, not just a Bluetooth logo.

USE-CASE HOOK: Keyboard, headset, and wearable nearby? Think Bluetooth. Internet over a local wireless LAN? Think Wi-Fi. Their radios may share a band, but the technologies differ.

Bluetooth pairing: trust before use

VISUAL: Draw a phone and headphone character comparing matching glowing symbols before exchanging sound notes.



Match the code. Then test the sound!

MATCH!

- **Pairing.** The process used to establish a trusted relationship and keys between devices. The steps depend on the accessory and operating system.
- **Discovery.** A device makes itself findable as required. Some accessories need a pairing-button press and a limited pairing window.
- **PIN / passkey.** PIN means Personal Identification Number. Enter or confirm the displayed code when the method requires it. Check that the intended devices show matching information.

- **Pairing versus connecting.** A remembered pairing can exist while the device is disconnected. The accessory may be off, out of range, or connected to another host.
- **Troubleshoot.** Check power, charge, radio settings, distance, pairing mode, and existing connections. Remove and recreate the pairing when appropriate, then test the needed function.
- **Secure behavior.** Reject unexpected pairing requests. Leave discovery enabled only as needed, and keep device software current. Follow workplace policy. [1,19]

A+ SEQUENCE: Enable Bluetooth, enable pairing, find the device, enter or confirm the appropriate code, and test connectivity. The exact screens vary by device.

NFC & RFID: taps and tagged objects

VISUAL: Draw a phone almost touching a payment reader beside a separate warehouse arch reading a tagged package.



TAP! A tiny gap can carry useful data!

TAP!

- **NFC.** Near Field Communication. A radio technology at 13.56 MHz for very close communication. Common uses include contactless payments, tags, and quick device interactions. [20]

- **NFC distance.** Think a tap or near touch. The NFC Forum describes typical operation up to about 0.8 inches. Hardware and use case affect reach; this is not an exact universal cutoff.

- **RFID.** Radio-Frequency Identification. Readers communicate with tags to identify or track objects. Examples include inventory, badges, and tagged equipment. [21]

- **Passive tag.** A typical passive tag gets operating energy from the reader's field and returns information. It does not need its own battery for that basic job.

- **Active tag.** Uses a transmitter and usually its own power source. RFID also includes battery-assisted passive designs. Range depends on the tag family and system. [22]

- **Relationship.** NFC is related to high-frequency RFID, but not every RFID reader or tag supports NFC. RFID can use different frequency bands and much longer distances.

MEMORY HOOK: NFC = near-touch interaction. RFID = identify tagged objects. Bluetooth = nearby device links. Wi-Fi = local wireless networking. Match the job before the buzzword.

Wireless fault detective

VISUAL: Draw a robot detective comparing a tablet's readings in a clear room and behind a thick wall.



One clue. One test. Then the next clue!

TRACE!

- **Network missing.** Check Wi-Fi radio state, distance, SSID settings, supported band, country settings, and AP status. A 2.4-only client cannot discover a 6 GHz-only network.

- **Password rejected.** Check the intended SSID, passphrase, authentication mode, credentials, and saved profile. Strong signal cannot fix the wrong password.

- **Connected, no internet.** Separate local Wi-Fi from IP addressing, DNS, gateway, and ISP problems. Compare another device and test the local connection first.

- **Slow or unstable.** Check signal, congestion, retries, interference, channel width, client capability, and upstream limits. Compare near-AP and problem-location tests.

- **Drops while walking.** Check coverage gaps, roaming behavior, consistent configuration, and the client driver. More APs help only when planned and connected properly.

- **One accessory fails.** For Bluetooth, check charge, pairing, supported profile, and whether another host holds the connection. Test the actual audio or input function.

EXAM APPROACH: Use the scenario's evidence. A symptom can have several causes. Change one thing, verify the result, and document the fix. [1,11]

Flashcard training: say it before flipping

VISUAL: Draw a cyan AP teacher and purple headphone student using blank gold flashcards at a neon study desk.



Recall first. Reveal second. Repeat later!

REPEAT!

- **The interactive companion.** Open the PowerPoint in Slide Show mode. Choose a card from its menu. Say your answer aloud, then select **REVEAL ANSWER**.
- **Built-in controls.** Use **TRY AGAIN** to return to the same question, **NEXT CARD** to continue, and **CARD MENU** to choose a topic. The deck uses internal links, with no macros.
- **Honest recall.** Answer before looking. “I recognize that” feels easier than producing the answer yourself. Say the standard, band, and rate in the same order each time.

- **A repeatable routine.** Do a short round now. Revisit missed cards tomorrow, then again a few days later. Adjust the pace to your recall rather than rushing through every card.
- **Paper option.** The next six pages contain the same 24 question-and-answer pairs. Cover the answer column with paper. Reveal one answer after each attempt.
- **Progress.** Keep a tally on paper or in your notes. The PowerPoint provides navigation and answer reveals; it does not save scores or automatically schedule reviews.

HOW TO OPEN: Use PowerPoint Slide Show for the clickable deck. Editing mode or some preview apps may handle links differently. The PDF remains the complete reading and printable study guide.

Flashcards 01-04: cover, recall, reveal

●●● **VISUAL:** Draw the wireless team practicing with blank gold and blue cards under three bright stars.



Cover the right column. Say it. Check it!

FLIP!

QUESTION - recall first	ANSWER - reveal after trying
01. 802.11a Name its band and maximum PHY rate.	5 GHz; 54 Mb/s. A takes the 5 GHz road.
02. 802.11b Name its band and maximum PHY rate.	2.4 GHz; 11 Mb/s. B is the bicycle: 11.
03. 802.11g Name its band and maximum PHY rate.	2.4 GHz; 54 Mb/s. G shares B's band but A's top rate.
04. 802.11n Give the Wi-Fi generation, bands, and maximum PHY rate.	Wi-Fi 4; 2.4 and 5 GHz; up to 600 Mb/s. The maximum needs the top supported stream and width configuration.

SELF-CHECK: Mark each answer “got it” or “again” in your own notes. Revisit missed cards before your next full round. All speed ceilings are theoretical PHY rates.

Flashcards 05-08: cover, recall, reveal

●●● **VISUAL:** Draw the wireless team practicing with blank gold and blue cards under three bright stars.



Cover the right column. Say it. Check it!

FLIP!

QUESTION - recall first	ANSWER - reveal after trying
05. 802.11ac Give the Wi-Fi generation, band, and maximum PHY rate.	Wi-Fi 5; 5 GHz; about 6.9 Gb/s. AC operation is 5 GHz. Dual-band AC routers use another mode at 2.4 GHz.
06. 802.11ax Give the Wi-Fi generation, usual Wi-Fi 6 bands, and maximum PHY rate.	Wi-Fi 6; 2.4 and 5 GHz; about 9.6 Gb/s. 6E extends this technology into 6 GHz.
07. Wi-Fi 6E What does 6E add, and which IEEE standard does it use?	It adds 6 GHz operation using 802.11ax. Wi-Fi 6 alone does not promise a 6 GHz radio.
08. Wi-Fi 7 Name its IEEE standard and expand MLO.	802.11be; Multi-Link Operation. Extra context: up to 320 MHz in allowed 6 GHz operation.

SELF-CHECK: Mark each answer “got it” or “again” in your own notes. Revisit missed cards before your next full round. All speed ceilings are theoretical PHY rates.

Flashcards 09-12: cover, recall, reveal

●●● **VISUAL:** Draw the wireless team practicing with blank gold and blue cards under three bright stars.



Cover the right column. Say it. Check it!

FLIP!

QUESTION - recall first	ANSWER - reveal after trying
09. Band, channel, width Which is the neighborhood, street, and street width?	Band = neighborhood. Channel = street. Channel width = how much spectrum the street occupies. GHz labels common bands. MHz measures channel width.
10. 1, 6, 11 What does this memory hook describe?	The common U.S. nonoverlapping 2.4 GHz channel plan with 20 MHz channels. It is not a universal channel plan for 5 or 6 GHz.
11. Wider channels Why can 20 MHz work better than 40 MHz in a crowded 2.4 GHz area?	It occupies less spectrum, leaving more room for channel reuse and reducing overlap. A wider channel can raise the ceiling while worsening a crowded deployment.
12. DFS Expand DFS and name the signal that triggers avoidance.	Dynamic Frequency Selection; radar. Some 5 GHz channels require radar detection and channel-avoidance behavior.

SELF-CHECK: Mark each answer “got it” or “again” in your own notes. Revisit missed cards before your next full round. All speed ceilings are theoretical PHY rates.

Flashcards 13-16: cover, recall, reveal

●●● **VISUAL:** Draw the wireless team practicing with blank gold and blue cards under three bright stars.



Cover the right column. Say it. Check it!

FLIP!

QUESTION - recall first	ANSWER - reveal after trying
13. MIMO Expand MIMO. What does it add?	Multiple Input, Multiple Output; multiple transmit/receive paths that can support spatial streams. Both endpoints and the channel conditions affect the benefit.
14. OFDMA Expand OFDMA and explain its job.	Orthogonal Frequency Division Multiple Access; assigns channel resources to multiple users in one transmission. Picture different seats in one shared shuttle.
15. Speed versus frequency Does 5 GHz mean a 5 Gb/s download?	No. GHz measures radio frequency; Gb/s measures data rate. Useful throughput is also lower than the ideal PHY ceiling.
16. Wi-Fi range Does each Wi-Fi standard guarantee one maximum distance in feet?	No. Usable range depends on equipment, walls, interference, band, and the needed performance. Measure at the real site and test both directions.

SELF-CHECK: Mark each answer “got it” or “again” in your own notes. Revisit missed cards before your next full round. All speed ceilings are theoretical PHY rates.

Flashcards 17-20: cover, recall, reveal

●●● **VISUAL:** Draw the wireless team practicing with blank gold and blue cards under three bright stars.



Cover the right column. Say it. Check it!

FLIP!

QUESTION - recall first	ANSWER - reveal after trying
17. SSID Expand SSID and explain its job.	Service Set Identifier; the wireless network name. A hidden name does not provide meaningful access security.
18. Backhaul What does a mesh node's backhaul carry?	Traffic between the node and the upstream network. It can use Ethernet or a wireless link.
19. Wireless security Expand WPA3 and SAE.	Wi-Fi Protected Access 3; Simultaneous Authentication of Equals. WPA3-Personal uses SAE for password-based authentication.
20. Bluetooth Name its band and two common jobs.	2.4 GHz; examples include headphones, keyboards, mice, and sensors. About 33 ft is only a short-range example, not a universal limit.

SELF-CHECK: Mark each answer “got it” or “again” in your own notes. Revisit missed cards before your next full round. All speed ceilings are theoretical PHY rates.

Flashcards 21-24: cover, recall, reveal

●●● **VISUAL:** Draw the wireless team practicing with blank gold and blue cards under three bright stars.



Cover the right column. Say it. Check it!

FLIP!

QUESTION - recall first	ANSWER - reveal after trying
21. Bluetooth pairing What should you check before accepting a displayed pairing code?	Confirm it belongs to the intended devices and matches when the method requires it. Then test the actual function. A saved pairing does not prove an active connection.
22. NFC Expand NFC and give its frequency and typical distance idea.	Near Field Communication; 13.56 MHz; a tap or near touch, typically up to about 0.8 inches. Think contactless payment or a nearby tag.
23. RFID Expand RFID. How does a typical passive tag get power?	Radio-Frequency Identification; from the reader's radio field. Active tags use a transmitter and usually their own power source.
24. Connected, no internet Does a strong Wi-Fi icon prove the ISP connection works?	No. Test local connectivity, IP settings, DNS, gateway, and the ISP path separately. Wi-Fi is the local radio link. The internet path has more parts.

SELF-CHECK: Mark each answer “got it” or “again” in your own notes. Revisit missed cards before your next full round. All speed ceilings are theoretical PHY rates.

Sources & exam alignment

VISUAL: Draw the wireless team checking gold reference cards at a neat neon desk.



Facts first. Memory hooks help them stick!

CHECK!

Scope: Primary objective 2.2. Related support: mobile connectivity 1.2/1.3, network setup, and network troubleshooting. Security and advanced Wi-Fi 7 features are supporting context. [1]

1. [CompTIA: Core 1 exam objectives](#)
2. [Cisco: Wireless RF reference](#)
3. [Intel: Wi-Fi generation comparison](#)
4. [Cisco: Radio channel configuration](#)
5. [Intel: Wi-Fi 7 features and products](#)

6. [FCC: 6 GHz device classes and rules](#)
7. [Cisco: 802.11n rates and capabilities](#)
8. [Cisco: The move to 802.11ac](#)
9. [Intel: Wi-Fi technical tutorial](#)
10. [Cisco: Wi-Fi 6E and client connectivity](#)
11. [Cisco: Wireless speed troubleshooting](#)

Reviewed September 18, 2026. Independent study material, not an official or endorsed CompTIA or Per Scholas publication. Speeds are theoretical maximums unless stated otherwise.

Sources: setup and short-range wireless

VISUAL: Draw the wireless team checking gold reference cards at a neat neon desk.



Facts first. Memory hooks help them stick!

CHECK!

Source note: Numbered references in the lessons lead to these primary sources. Follow current local rules and exact device manuals for real installations.

12. Apple: [Wireless network planning](#)

13. Apple: [Router and AP settings](#)

14. Apple: [Wi-Fi privacy and identifiers](#)

15. Microsoft: [Wi-Fi Direct](#)

16. Cisco: [Wireless security](#)

17. Bluetooth SIG: [Technology overview](#)

18. Bluetooth SIG: [Understanding range](#)

19. Microsoft: [Pair a Bluetooth device](#)

20. NFC Forum: [NFC technology](#)

21. GS1: [RFID standards and uses](#)

22. GS1: [Passive and active RFID tags](#)

Original comic artwork created with built-in image generation. Characters and glowing radio paths are memory aids. Physical distances use inches and feet. The flashcard PowerPoint uses clickable links, not macros.

STAR QUIZ TIME

 **VISUAL:** Draw the wireless team celebrating under three gold stars with blank flashcards on a bright desk.



Three missions. You have the clues!

STAR!

- 1. The missing new band:** A laptop supports Wi-Fi 6 on 2.4 and 5 GHz. Can it join a 6 GHz-only network just because both names contain “6”? What extra support does it need?
- 2. The crowded classroom:** Three nearby APs use 20 MHz channels in the U.S. 2.4 GHz band. Which familiar three-channel plan avoids their channels overlapping? Does that guarantee zero interference?
- 3. The tap station:** A phone sits less than an inch from a payment reader. Which technology best fits: Wi-Fi, Bluetooth, or NFC? Expand its name and give its frequency.

ANSWERS - CHECK YOUR HERO SKILLS!

- 1. No. Great catch!** It needs a client radio and software supporting 6 GHz, such as a suitable Wi-Fi 6E or Wi-Fi 7 device, plus compatible security and local permission.
- 2. Channels 1, 6, and 11. Nice recall!** They are the common nonoverlapping plan for this setting. Other networks and non-Wi-Fi devices can still cause contention or interference.
- 3. NFC. You matched the job!** Near Field Communication operates at 13.56 MHz. Think a tap or near touch; actual range depends on the hardware and use case.